

Nandana Narayan Das

+91 7907382133 | nandanadas007@gmail.com | github.com/nandana-das | linkedin.com/in/nandana-narayan-das

PROFILE

M.Tech AI & Data Science student at Alliance University, Bangalore. I build things — a RAG system for ISRO domain knowledge, a computer vision pipeline for lunar surface detection, a multi-agent LLM system, and research published at IEEE. I'm comfortable working across the ML stack: data pipelines, model development, GenAI integrations, and explainability. Looking for an internship where I can contribute from day one.

TECHNICAL SKILLS

LLM / GenAI: LangChain, LangGraph, RAG, Knowledge Graph RAG, FAISS, prompt engineering

Computer Vision: OpenCV, YOLO, RT-DETR, CNN, PyTorch, TensorFlow, image classification, object detection

ML / DL: scikit-learn, Keras, LSTM, Transformer, CNN-LSTM, NLTK

Explainable AI: SHAP, Grad-CAM, LIME

Data & Analytics: Pandas, NumPy, SQL, Power BI, Matplotlib, ETL, EDA, Streamlit, spaCy

Languages & Tools: Python, Java, C, SQL, Git, GitHub, Jupyter Notebook, Google Colab

PROJECTS

KG-RAG-ISRO | *Python · LangChain · FAISS · spaCy · Knowledge Graphs · Streamlit*

- Built a Knowledge Graph-augmented RAG system for ISRO domain Q&A — combining graph-based retrieval, FAISS vector search, spaCy entity extraction, and local LLM inference into an interactive Streamlit app.
- Goes beyond standard RAG: structured entity-relationship retrieval from a knowledge graph improves answer grounding and factual accuracy on domain-specific queries.

Lunar Landslide & Boulder Detection | *Python · YOLO · RT-DETR · PyTorch · OHRC Imagery*

- Built an object detection pipeline using YOLO and RT-DETR to detect landslides and boulders in ISRO's Chandrayaan OHRC imagery — a domain where labeled data is scarce and standard augmentation strategies must be rethought.
- Implemented full pipeline: label conversion, multi-stage training, inference, confidence analysis, and architecture comparison across detection models.

AI Resume Agent | *Python · LangChain · LangGraph · Groq · Gemini API*

- Designed a four-agent LangGraph pipeline — JD parser, resume evaluator, gap analyst, resume writer — on a shared state graph, auto-generating tailored resumes from job descriptions.
- Integrated Groq and Gemini APIs with structured prompt engineering and output validation; built a companion Study Buddy Streamlit app on Gemini API.

IGP Emissions Attribution | *TensorFlow · CNN · LSTM · Transformer · SHAP · Grad-CAM*

- Hybrid CNN-LSTM-Transformer model fusing satellite imagery and meteorological time-series for air pollution source attribution over Delhi NCR; F1-score 0.93 for emission detection, 0.76 for source attribution.
- Applied SHAP and Grad-CAM for model transparency and bias detection. Presented at conference — publication under review.

Smart City Traffic AI — Bengaluru | *Python · LSTM · TomTom API · Fuzzy Cognitive Maps · Flask*

- Full-stack traffic forecasting system: LSTM congestion prediction on live TomTom API data, Fuzzy Cognitive Map explainability, Flask backend, and separate frontend — built as a team project.

PUBLICATIONS

Explainable AI for Disease Detection and Body Constitution Analysis Using Tongue Imaging — IEEE Xplore, ICTEST 2025 [View on IEEE](#)
Physics-Informed Explainable Multimodal Deep Learning for Satellite-Based Air Pollution Source Attribution in Delhi NCR — *Presented at Conference, Publication Under Review*

CERTIFICATIONS

Oracle Certified Java Foundations Associate · IBM AI Foundation · Microsoft Generative AI Essentials · GitHub Professional Certificate · KNIME Data Science Certificate · Infosys AI Foundation

EDUCATION

M.Tech in Artificial Intelligence and Data Science 2025 – 2027

Alliance School of Advanced Computing, Alliance University — Bangalore, Karnataka

B.Tech in Computer Science and Engineering (AI) 2021 – 2025

Adi Shankara Institute of Engineering and Technology — Kalady, Kerala